Machine Learning Engineer

Vedra AI · Toronto, hybrid · Full-time

About Vedra AI

Behind every claim there is an adjuster deciding what happens next with fragmented information and limited time. Vedra AI exists to make those decisions faster, fairer, and better informed.

We're building insurance's first AI orchestration platform: the layer that brings together specialist AI for damage, injury, and fraud, and turns it into one clear, auditable recommendation inside the adjuster's workflow. It's live with Canadian carriers today, running inside Guidewire ClaimCenter.

We're a Toronto team of insurance and technology experts, and we're building our ML function now.

The role

This is a role for a recent graduate who wants their first models to matter. You'll work on the scoring models behind our claims recommendations, including P-negative, which estimates shortly after a claim is reported whether it's heading for an adverse outcome.

Level	Junior / new grad (0–2 years)
Team	AI & data science
Reports to	Head of AI
Location	Toronto, hybrid

What you'll do

- Build the models that help adjusters decide what happens next on a claim, from features through training, calibration, and evaluation.
- Turn messy carrier and third-party data into training sets you can stand behind, catching leakage, label bias, and gaps. Our rule: absent is never negative.
- Design evaluations that tell the truth: clean splits, thresholds set without peeking at test data, and metrics that reflect how adjusters actually use a score.
- Close the loop: use adjuster overrides and claim outcomes to make each model better than the last.
- Explain your work in plain language to carriers, compliance teams, and colleagues outside ML.
- Explore how signals from multiple specialist AI vendors combine into better decisions, a problem that's still largely unsolved across the industry.

What you'll bring

We care more about how you think than where you studied. You'll do well here if you have:

- Strong Python and hands-on experience with pandas, NumPy, and scikit-learn; some exposure to PyTorch or a gradient-boosting library such as XGBoost or LightGBM.
- Strong grounding in probability, statistics, and core ML: classification, overfitting, cross-validation, class imbalance, and calibration.
- At least one end-to-end ML project (course, thesis, internship, co-op, or personal) you can walk us through, including what went wrong.
- Clear written communication and the curiosity to learn an unfamiliar domain like insurance claims.

Nice to have

- A bachelor's or master's degree in computer science, statistics, mathematics, engineering, or a related field, completed in the last two years or finishing by your start date.
- Working SQL and comfort with messy, real-world tabular data.
- Exposure to fraud detection, risk scoring, survival analysis, or other imbalanced-outcome problems.
- Experience with model explainability.
- Familiarity with cloud tooling (AWS preferred), Docker, Git, and PostgreSQL.
- Experience working with LLMs or building on top of third-party AI APIs.
- Interest in insurance, financial services, or regulated AI.

How to apply

Send me your resume.